

Complex mathematical formulas

MATH-P1

Display math mixes integrals, limits, radicals, matrices, sums, and Greek symbols.

Eq. 1

$$F(k) = \int_{-\infty}^{+\infty} e^{-i2\pi kx} f(x) dx$$

with $k \in \mathbb{R}$ and $i^2 = -1$

Eq. 2

$$\lim_{n \rightarrow \infty} \sum_{j=1}^n \frac{(-1)^j \cdot j^2}{n^3 + j} = -\frac{1}{12}$$

Normalized: $\lim_{n \rightarrow \infty} \sum_{j=1}^n ((-1)^j j^2) / (n^3 + j) = -1/12$

Matrices and partial differential equation MATH-P2

Exact normalized strings are stored in ground_truth.json.

Eq. 3

$$A = \begin{bmatrix} \alpha & \beta^2 & \sqrt{\gamma} \\ 0 & e^{i\theta} & -\lambda \\ \partial x & \partial y & 1/2 \end{bmatrix} \quad \det(A) \neq 0$$

Eq. 4

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} = -\nabla p + \nu \nabla^2 \mathbf{u}$$

subject to $\nabla \cdot \mathbf{u} = 0$